

Repository 6:

Automation and Wages

$$i \in [N-1, N]$$

$$I^* = \min\{I, \tilde{I}\}$$

$i \leq I^* \Rightarrow$ task performed by capital
 $i > I^* \Rightarrow$ task performed by labor

Technology - constrained regime:

$$I^* = I < \tilde{I}$$

An increase in I expands the set of tasks performed by capital.

$$\frac{d \ln(W/R)}{dI} = - \frac{\Delta_I}{\Theta + \epsilon_L} < 0$$

Therefore,

$$dI > 0 \Rightarrow d \ln(W/R) < 0$$

Displacement effect: automation reallocates tasks away from labor, reducing labor's relative demand

But a fall in W/R does not necessarily imply a fall in W

$$d \ln W = \underbrace{d \ln Y|_{K,L}}_{\text{Productivity effect (+)}} + \underbrace{(1-S_L) d \ln(W/R)}_{\text{Displacement effect (-)}}$$

Using the automation effect:

$$d \ln W = d \ln Y|_{K,L} - (1-S_L) \frac{\Delta_I}{\Theta + \epsilon_L} dI$$

Hence,

$$dW > 0 \Leftrightarrow d \ln Y|_{K,L} > (1-S_L) \frac{\Delta_I}{\Theta + \epsilon_L} dI$$

So:

Productivity effect > Displacement effect $\rightarrow W \uparrow$
Productivity effect < Displacement effect $\rightarrow W \downarrow$

Proposition 3:

$$\exists \bar{K} < \infty$$

Such that

$$K > \bar{K} \rightarrow dW/dI > 0$$

and

$$K < \bar{K} \rightarrow dW/dI < 0$$

Verdict

Automation unambiguously lowers w/r in the technology-constrained regime, but it does not necessarily lower the real wage. The sign of the wage effect depends on whether productivity gains dominate displacement.

Even if

$$W \uparrow,$$

Automation can still imply

labor share $\downarrow$, employment $\downarrow$